



NewBear Computing Store Ltd



HEAD OFFICE: 40 BARTHOLOMEW ST. NEWBURY BERKS. RG14 5LL Tel: (0635) 30505 Telex: 848507NCS

DESIGN NOTE 38

ROM MONITOR BOARD FOR 77-68

This PCB has the following features:

1. Buffered address and data bus.
2. Mikbug, SWATBUG monitor with PIA for serial interface.
3. Timer for the serial interface with preset speeds up to 1200 b/s.
4. 1K of EPROM (2708) positioned at E400₁₆ - E7FF₁₆
5. PIA for parallel interfacing situated at 8010₁₆ - 8013₁₆
6. ACIA for serial interfacing at 8000₁₆ - 8001₁₆
7. Single step facility - detects E112₁₆ (RT1) and operates NMI to save the register contents after each program step. To allow proper functioning of the serial I/O the interrupt is held while addressing 8000₁₆ and above. (MIKBUG only).
8. The least three bits of the top address are repeated down to FC00 and read from a 32 x 8 PROM giving return addresses for the various interrupts. The remaining 2 address bits can be used to provide 4 sets of these addresses.
9. Existing 77/68 facilities can be retained.

Any errors in this design note can be found in the Summer '79' User Group Newsletter or from Newbear on request.

Introduction

MONITORS

77/68 constructors will all be aware that entering all but the smallest programs into memory is a long, tedious and error prone process. Debugging is also far from easy. Manufacturers have long recognised this problem and have attempted to alleviate it by supplying various 'monitor programs' normally in read only memory. Monitor programs are simple programs which allow some sort of keyboard or keypad to be used for input and also allow output to a printer or other type of device. They contain code to load and examine memory and also to enter a program. Most monitors presuppose a minimal amount of hardware of a very simple form. The early monitors are very primitive but later ones have various extra facilities such as break point features and code for driving more exotic hardware.

The M6800 has a number of monitors both 'soft' ones in RAM and 'hard' ones in ROM. The soft monitors have the disadvantage that they have to be loaded into ram each time the system is powered up. On the other hand they have the advantage that they can be enhanced and altered as required. Code is often designed to run under a specific monitor in that it takes advantage of some of the routines in the monitor particularly for input/output. The 77/68 manual has a design for a soft monitor board and the listing of a suitable monitor. Of course the code from one of the hard monitors can be cribbed and loaded into a soft monitor as long as the hardware is suitable. In the case of the MON 1 board this is not possible without hardware changes as the memory map is different to that of the popular hard monitors.

There are at least five hard monitors commercially available with the M6800 and probable many more. The hardware of the MON 2 board will support MIKBUG (a MOTOROLA offering) or a number of so called MIKBUG compatible monitors such as SWTBUG or R768. MIKBUG is 512 bytes in a 1K ROM. The other 512 bytes are taken up by MINIBUG which is 256 bytes and lacks the facility to output store in a form for reinput and also uses different hardware to MIKBUG. The remaining 256 bytes is another copy of MINIBUG. Another well known monitor is JBUG but this is designed for the D2 kit and uses a HEX keypad for input and a small LED display for output. MIKBUG uses an asynchronous (teletype) interface and is not very useful unless the constructor has such a device to hand.

SWTBUG offers all the facilities of MIKBUG and can operate with either asynchronous interfaces based on a PIA (as used by MIKBUG) or ones based on an ACIA which are rather superior in the signals they can generate to the outside world. In addition up to 8 interfaces can be utilised which is convenient for dedicating interfaces for various purposes such as a cassette interface. SWTBUG also includes code to load from a disc as long as the disc is built to SWTBUG standards. A useful breakpoint facility is included so that a program on reaching given points in the code control is returned to the monitor to allow the central registers or store to be examined.

Care has been taken to ensure that important routines in the MIKBUG compatible monitors start at the same addresses so that code designed to run under MIKBUG, which uses this code, will work. Unfortunately some code may use less important routines which do start at different addresses. To help solve such problems SWTBUG has a byte search command which allows a given area of store to be searched for a given byte which could be a routine call. In this way code should not be too difficult to alter to run under SWTBUG.

A disappointing feature of MIKBUG and the MIKBUG compatible monitors is the way the various areas of store required are scattered randomly over the top half of the memory which could be embarrassing on a large system.

There seems little point in using MINIBUG since little software exists to run under it and moreover its facilities are very restricted. JBUG, as far as 77/68 is concerned, is similarly endowed. Since most software which will run under MIKBUG will also run under SWTBUG, with only minor changes, and as SWTBUG has superior facilities there seems little reason for selecting a MIKBUG monitor. The choice is therefore between SWTBUG and a soft monitor. The choice depends on the use to which a system is to be put. If existing software is to be run SWTBUG is far more useful. If the interest is in writing monitors or driving novel devices then the soft monitor has a lot to recommend it.

References

Constructors are strongly recommended to obtain the manuals corresponding to the monitors they wish to use as these contain full listings of the monitors and details of use.

1. SWTBUG 6800 ROM MONITOR VERSION 1.0 USERS GUIDE
Published - Southwest Technical Products Corporation.
2. MCM6830L7 MIKBUG/MINIBUG ROM ENGINEERING NOTE 100.
Published - Motorola Semiconductor Products Inc.
3. 77/68 MON 1 BOARD DESIGN NOTE 22
Published - Newbear.

THE ROM MONITOR

The ROM MONITOR has been designed to interface with the 77/68 CPU in order to extend its facilities and application. It makes available to the user a Monitor programme in ROM, either MIKBUG or the updated version SWATBUG. Alternatively the electrically alterable PROM 2708 can give the user the option of employing a customised version to suit his particular applications, or it may simply be used to store some frequently used programme.

The Monitor greatly relieves the monotony of loading programs by switches. This is amply illustrated by the memory change facility offered by SWATBUG. With this device, once the initial address has been entered the program may be typed in without further reference to the address or to any special control characters. Another feature is its ability to decode and generate the ASCII character set, to give easy access to a keyboard, VDU, cassette or tape punch via a standard V24/RS232 datalink.

For the complete MIKBUG/SWATBUG facilities see the relevant application notes.

To aid programme development a single-step facility has been included, with this arrangement a programme cannot run berserk. After each instruction is executed the programme counter is held and the internal registers are displayed for diagnostics.

An additional PIA is included giving 16 I/O ports for monitoring and or parallel tape loading etc.

The lack of instruments can make testing very difficult. Here, some thought has been given to this aspect and the system has been designed so that the board can be checked during its assembly by a series of test programmes. In fact the only instrument required should be a multimeter.

The ROM MONITOR has many other features which will be described in the notes that follow.

Figure 1 is a functional block diagram for the board. This and the full circuit diagram may be used as background to the technical description that follows:

The main features of the ROM MONITOR PCB assembly are as follows:

- * Maintains all the facilities of the 77/68 CPU.
- * Buffered address and data lines.
- * MIKBUG/SWATBUG monitor ROM.
- * Programmable timer for V24 serial interfacing. Signalling speeds can be preset to 110, 300 or 1200 baud.
- * PIA used by MIKBUG/SWATBUG for serial interfacing.
- * ACIA at address 8000₁₆ for user programming; can be moved to 8004₁₆ for high speed serial interfacing with SWATBUG.
- * Access to the 256 bytes of scratch-pad memory located on the 77/68 CPU PCB at base address A000₁₆.
- * Single step facility which can display the contents of the CPU's registers after each programme step.
- * 1K byte of EPROM located at base address E400₁₆

- * Reset/interrupt vectoring. The least three bits of the top address read from a 32 x 8 bit PROM which has been programmed with four sets of 8 vectors. Each set can be associated with one of the following MIKBUG/SWATBUG EPROM, Scratch-pad or main memory.

Technical Description

The address lines are buffered by X2 and X5. They must be the non-inverting type to allow the monitor ROM and PIA's to function correctly. The ROM/PROM is write protected by X1(a) which pulls the address line BA14 low during a write operation, thereby inhibiting the chip enable signals for X20 and X7.

Parts b, c and d of X1 buffer the R/W, Ø2 and VMA lines respectively, they are also used to effectively remove the monitor board from the bus to allow reversion to the original basic 77/68 facilities.

All the chip enable signals are decoded by X9/12, 25, 21, 22, 16 and 17. These are as follows:

- CE2 Taken off the PCB to enable the scratch-pad memory.
- CE1 Enables the PROM X20 which holds the reset/interrupt vectors.
- CE3 Decodes E000₁₆ - E3FF₁₆ to enable MIKBUG.
- CE4. CE5 Decodes 8004₁₆ MIKBUG PIA position.
- CE4. CE6 Decodes address 8010₁₆ user PIA position.
- CE4. CE8 Decodes address 8000₁₆ - 8001₁₆ user ACIA position.
- CE9 Decodes E400₁₆ - E7FF₁₆ EPROM position.

Parts of X26, 22 and 23 performs a logical 'OR' function on all the chip enables except CE2 (scratch-pad enable) and CE7 (MIKBUG RT 1) then the result is 'AND'ed in X23 with WE to enable the read data buffers X13, 14. Do note that 16 also enables the buffers for the address 8014₁₆ 8017₁₆ which is not occupied at the present. The write buffers (part off X13, 14) are enabled by the read/write control WE. For the complete memory map see Figure 2.

The internal registers of PIA's X8 and X4 are initialised by RESET which enters the board via edge connector pin 73 from the processor PCB. The ACIA has an internal power up reset.

Line Driver X18 and receiver X19 convert the PIA's TTL input/output ports PA7, PA0 respectively to V24/RS232 specified line voltages. The CB2 output is buffered via the line drive X18 and may be used for tape reader control and/or to disable via gates the PIA port echo.

Timer X24 is reset by PB0, (pin 10, X8) and half/full width timing is controlled by PB2 (pin 12, X8). The timing pulse is input via PB7 (pin 17, X8). The programmable divider controls with presets R16, 17 and 18 are brought to switches X13. The component values are such that three baud rates can be selected, they are 110, 300 and 1200 baud. The switch functions are given in Figure 3.

The ACIA data input/output and control lines are taken out to edge connector pins. Although the PIA at address 8004₁₆ will work into SWATBUG or MIKBUG. SWATBUG will automatically work with either a PIA or ACIA. If the ACIA is preferred simply interchange chip signals CE5 and CE8 to reposition the ACIA at 8004₁₆.

The EPROM X7 has its -5v supply derived on board from the -12v rail. The chip enable decoder locates it at address E400₁₆ - E7FF₁₆. However address bits A12, A12 and A11, A11 are available hence some rerouting of the inputs to pins 3, 4 and 5 of X10 will relocate the EPROM anywhere within a 4K byte address space upwards from E000₁₆.

Single Step Facility

With the MIKBUG ROM the user runs a programme by placing the start address in the stack A048₁₆ and A049₁₆ then typing 'G'. This will call the RT1 instructions at E112₁₆ which puts the contents of the stack into the CPU registers.

If ESS (enable single step) is high E112₁₆ on the address bus will cause CE7 to go low loading the 4 bit counter X28 with 15₁₀, clearing X30 and setting X29 'Q' output high.

A total of 10 Ø2 cycles later, X29 pin 11 will go high clearing X29 and disabling X27 (a) and NMI. One half cycle later, if BA15 is low X29 pin 5 will go high enabling X27 (a) and allowing NMI to go low. This cycle will be repeated each time CE7 goes low in response to address E112 on the bus. Timing waveforms are given in Figure 4.

RT1 required eleven Ø2 cycles to complete, NMI will go low at the beginning of the eleventh, which is the fetch instruction cycle of the next programme step. Since it occurs less than 200ns before Ø1 this instruction will be completed before the registers are saved in the stack.

Next, the interrupt service routine whose address is recovered from scratch-pad at A006/7₁₆, will save the stack pointer and then cause a jump to the register dump programme in MIKBUG. Typing 'G' will repeat the whole sequence for the next programme step.

The following is suitable data to operate the single-step facility.

ADDRESS	DATA
A006	A0
A007	4A
A04A	BF
A04B	A0
A04C	08
A04D	7E
A04E	E1
A04F	1F

Linking the ROM MONITOR to the CPU Board

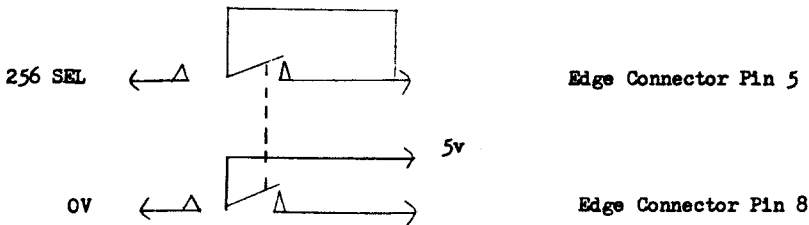
Here it is suggested that the switches and LED's should remain, they will prove useful for general fault finding and testing. If a rack is to be used control could be transferred to a front panel using dual in line switches instead of single toggles.

Certain modifications to the 77/68 board are required these are:

1. Strap X24 Pin 8 to the edge connector, pin 73.
2. Replace R4 430 with 1K
3. An additional double pole changeover switch if connected as shown below will allow the selection of either the ROM MONITOR or the existing 77/68 facilities.

77/68

ROM MONITOR



Construction

- Make all wire links
- Insert all capacitors and discrete components
- Check for supply shorts
- Check for correct supplies at each IC position
- Fit ribbon cable
- Insert only the following devices using the minimum of solder to avoid bridging pin and track.

X5, 6 9/12, 21/23, 25, 30, 16, 17, 1 and X2.

- Connect to 77/68 bus, make sure the unused pins are disconnected and switch on power. See back wiring lists Figures 5 & 6.
- Set switch to 77/68.

The following test programmes can be used to check all the chip enable signals. The programme is entered as normal through the 77/68 front panel.

TEST 1CHECKS CE1

<u>Address</u>	<u>Data</u>	
00	7E	JMP QFF00
01	FF	
02	00	

Results

Volts of CE1 when reset 4v
Volts of CE1 when running 1.4v

TEST 2CHECKS CE2

<u>Address</u>	<u>Data</u>	
00	7E	JMP @ A000
01	A0	
02	00	

Results

Volts on CE2 when reset 0V
Volts on CE2 when running 2.4v

TEST 3CHECKS CE3

<u>Address</u>	<u>Data</u>	
00	7E	JMP @ E000
01	E0	
02	00	

Results

Volts on CE3 when reset 0.2v
Volts on CE3 when running 2.4v

TEST 4CHECKS CE4

<u>Address</u>	<u>Data</u>	
00	7E	JMP @ 8000
01	80	
02	00	

Results

Volts on CE4 when reset 0V
Volts on CE4 when running 2.4v

TEST 5CHECKS CE5

<u>Address</u>	<u>Data</u>
00	01 NOP
01	01 NOP
02	01 NOP
03	01 NOP
04	7E JMP 0004

<u>Address</u>	<u>Data</u>
05	00
06	04

Results

Volts on CE5 when reset 0V
Volts on CE5 when running 2.4v

TEST 6

CHECKS CE6 and CE8

<u>Address</u>	<u>Data</u>
00	7E JMP @ 0000
01	00
02	00

Results

Volts on CE6 when reset 4v
Volts on CE8 when reset 0v
Volts on CE6 when running 0.4v
Volts on CE8 when running 2.4v

TEST 7

CHECKS CE7

<u>Address</u>	<u>Data</u>
00	7E JMP 0012
01	00
02	12 A JMP @ E112
12	7E
13	E1
14	12

Results

Volts CE7 when reset 4v
Volts CE7 when running 3v

TEST 8

CHECKS CE9

<u>Address</u>	<u>Data</u>
00	7E JMP @ E400
01	E4
02	00

Results

Volts on CE9 when reset 4v
Volts on CE9 when running 1.4v

During tests 1, 3 and 8 check that X23 Pin 8 = 2.4 volts running
= 0 volts reset

All test voltages were measured using a multimeter greater than 20,000^Ω/v.

After these tests have been completed set switch X31 to quadrant 1 (see figures 3 & 7). This will give a start address at A050₁₆ after reset.

Insert IC's X20, 13 and 14 then with ESS low a programme can be loaded starting at address 50₁₆. A suitable test programme would be as follows:

<u>Address</u>	<u>Data</u>
50	B6
51	A0 Load ACC A with contents of A060
52	60
53	B7
54	A0 Store ACCA in switch Register (LED's)
55	FF
60	11

Set switch X31 to ROM MONITOR press RESET and set CPU to RUN. If everything is correct 11₁₆ should appear on the LED display. Note that the programme can be started with the data switches in any position, the starting address A050₁₆ has been obtained from ROM X20.

Almost everything has now been checked, all that remains is to plug in the monitor ROM and adjust the timer to give the right BAUD rate.

Now, observing the usual precautions when handling MOS devices plug in the remaining IC's. Be careful - MIKBUG socket is aligned differently to all the other IC's check against the layout diagram.

Set switch X31 for the designed BAUD rate and to select the first quadrant (see figures 3 & 7) Press RESET repeatedly while adjusting the appropriate preset (figure 3) until a NEW LINE * response is obtained.

SWATBUG

The procedure is identical except that quadrant 2 is used and the response is NEW LINE §.

EPROM

Before inserting check voltage supplied are correct.

That concludes the assembly and test. Three simple programmes that will exercise most of the system are given on the coding form that follows. After the programme has been entered via MIKBUG/SWATBUG and the appropriate starting address selected the display (LED's) will increment sequentially, randomly or sequentially but at a speed controlled by the data switch position.

See 77-68 manual for further information.

24 COLLEGE ROAD,
MAIDENHEAD,
BERKSHIRE
SL6 6BN.

8 BIT MICROCOMPUTER		TYPE	3800			
PROGRAM ROM MON TEST			ISSUE	1	AUTHOR	DATE July 78
					PAGE	1 OF 1
ADDRESS	MACHINE CODE	LABEL	HEXHEMONIC	COMMENTS		
				DISPLAY COUNTS UP IN BINARY		
A04A	08	START	INX			
4B26	FD		BNE START			
4D4C			INC A			
4E87	A0FF		STA A 00FF			
5120	F7		BRA START			
			END			
				SIMPLE RANDOM SEQUENCE ON DISPLAY		
5308		START	INX			
5426	FD		BNE START			
564C			INC A			
5749			ROL A			
5887	A0FF		STA A 00FF			
5B20	F6		BRA START			
			END			
				VARIABLE SPEED COUNT —		
5D5F		START	CLR B	CONTROLLED BY SW REG.		
5E4C		LOOP	INC A			
5F26	FD		BNE LOOP			
615C			INCB			
62F1	A0FF		CMPB A0FF			
6526	F7		BNE LOOP			
6708			INX			
68FF	A0FE		STA A 0FE			
6B20	ED		BRA START			
			END.			

Fallback Mode

If the address space above MIKBUG, including the EPROM X7, is not used the PROM X20 which sits at the top address may be made redundant. This can be achieved by cutting the track to X17 pin 9 and connecting pin 9 to 0V.

Rack System

If a Rack System is to be employed to hold you expanding system then a front panel that will hold all the controls and indicators might be considered. A suggestion is to attach it to a prototyping board that will give access to the rack wiring. A suitable panel layout is given in Figure 9 and will occupy 5 slots in a 5 u rack.

25 pin D type socket could be mounted in front panel.

Using the MIKBUG ROM for the first time

Press Reset * should be returned.

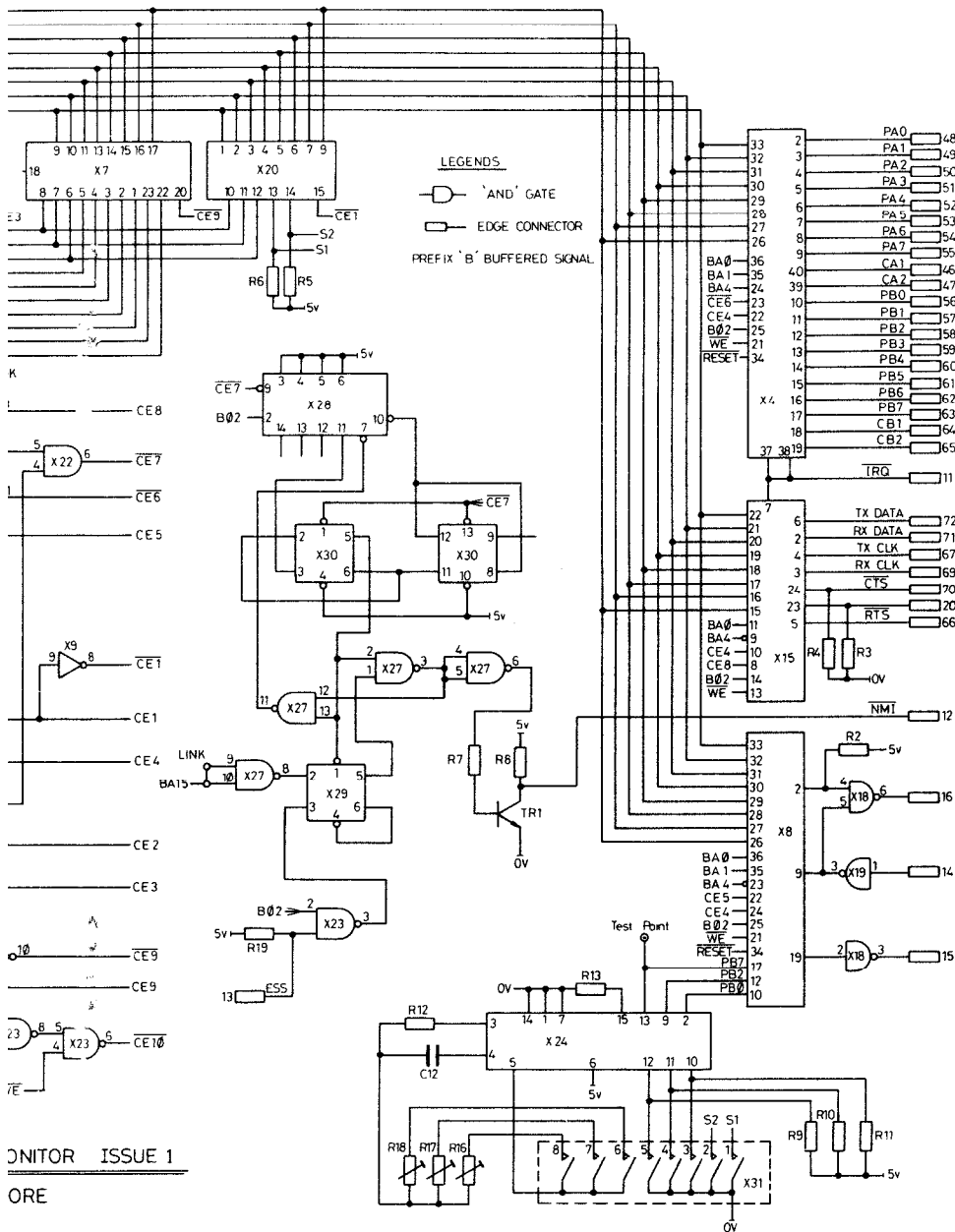
Type 'M' a space will be returned, now input the programme start address which will be echoed and its present contents output.

Input a space character then the first byte of your programme. In return the next address will be output with its present contents. Type 'space' then the next programme byte and so on until a complete programme has been entered. Exit from this routine by typing 'space' twice.

Next enter the start address in memory location A048 and A049, using the same technique. Then exit (type 'space' twice) and type 'G' to run the programme.

If your programme does not return to MIKBUG you can recover by pressing RESET.





DNITOR ISSUE 1
ORE

Various Modifications on the system when using SWTBUG

X15 Pin 9 and X4 Pin 24 taken from BA4, to change over the PIA-ACIA Port numbers (now are PIA=8000-8003 and ACIA=8010-8011). This is because SWTBUG's port 0 is written for PIA only. (Load & Punch routine).

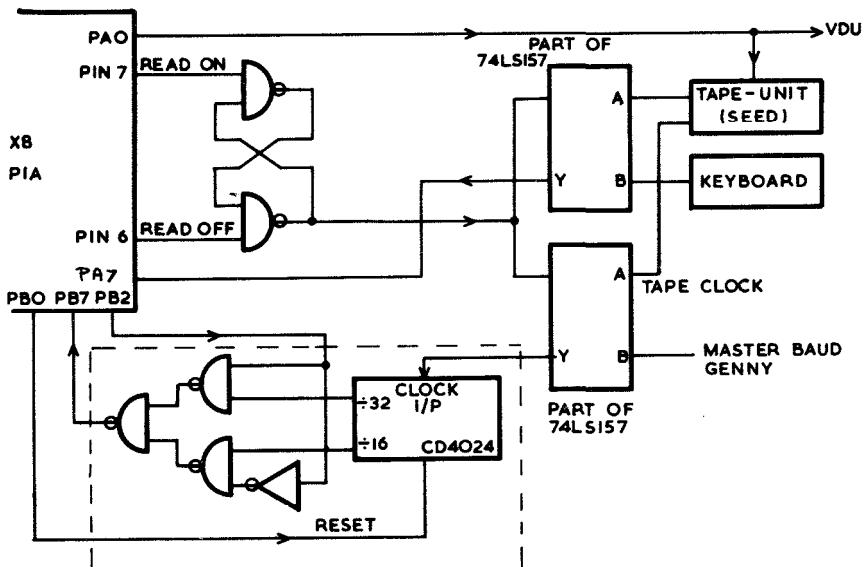
The beauty of this board is the ability to change parts of the circuit without too much trouble i.e. = by cutting track to Pin 9 of X17 and grounding Pin 9, use SWTBUG's top of addresses (X20 and gates for CE1 decoding removed). Lastly change the inputs to X10 Pins 3, 4 and 5 so that CE9 = C000 (HEX) which is entered when using SWTBUG by typing Z.

SWTBUG users???? You can't use SWTPC programmes??????

On SWTPC programs change the following RAM locations:

A00A on MIKBUG is	A00F on SWTBUG
0B is	A047
0C is	A00D
0D is	A00E
0E is	A046
A00F is	A044
10 is	A045
11 is	A047
12 is	A010
13 is	A011
A014 RMB 46	A017 RMB 43

Also using the J (ADDR) to jump to address to start your program, your first instruction is to redefine your stack or your random generator located in RAM is written into, other than these points SWTBUG is far better than MIKBUG.



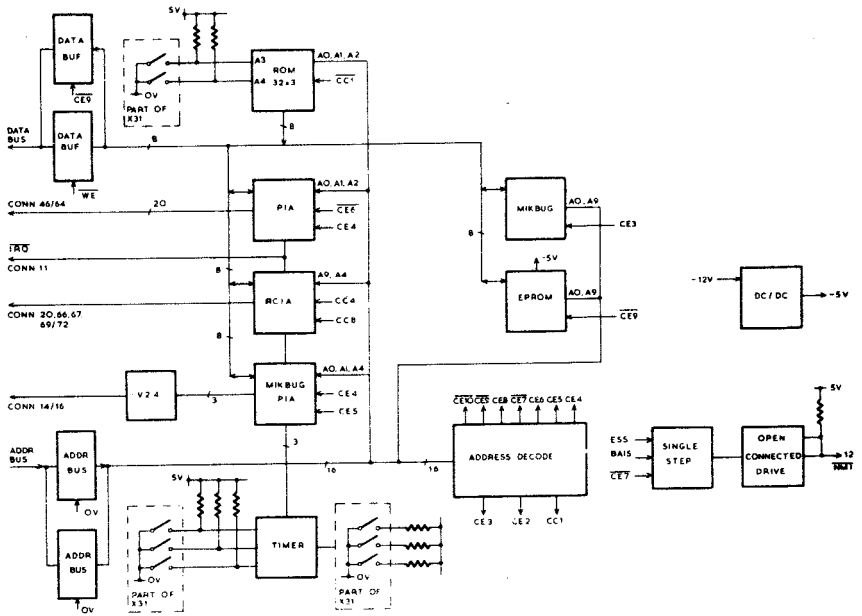


New Bear Computing Store
© 1979 ISS I



ROM MONITOR 77/78

FIG. 1



ROM MONITOR 16/6/78

Serial Data Rate & Interrupt Selection.

Switch

Dual in line Package.

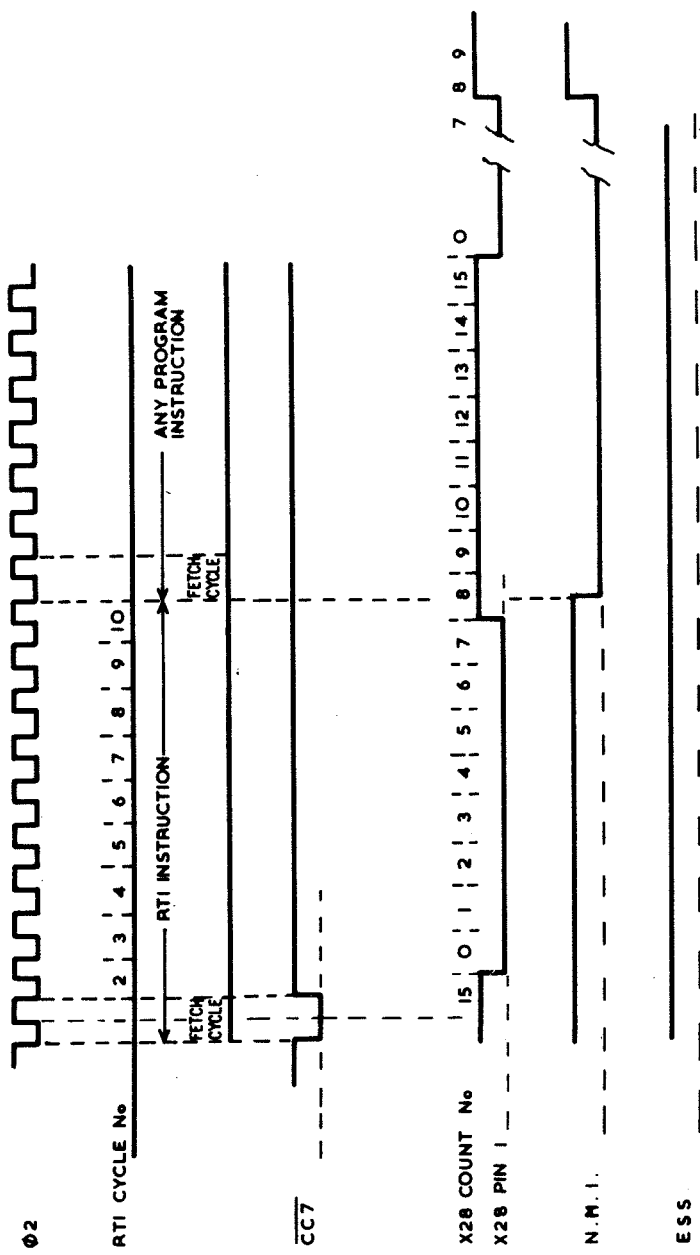
1	2	3	4	5	6	7	8	SWITCH NO.
****	****	****	****	****	****	****	****	'ON POSITION'
***	***	***	***	***	***	***	***	
**	**	**	**	**	**	**	**	
*	*	*	*	*	*	*	*	
----	----	----	----	----	----	----	----	'OFF POSITION'

	SWITCH NO.	1	2	3	4	5	6	7	8
<u>BAUD</u>									
110					ON.		ON.		
300				ON.	ON.			ON	
1200						ON			ON
ROM QUADRANT 1		ON	ON						
ROM QUADRANT 2		ON	OFF						
ROM QUADRANT 3		OFF	ON						
ROM QUADRANT 4		OFF	OFF						

CONTENTS.

ROM QUADRANT 1 .. Scratch Pad Addresses.
ROM QUADRANT 2 .. SWATEBUG Addresses.
ROM QUADRANT 3 .. SPARE
ROM QUADRANT 4 .. MIKBUG Addresses.

Figure 3



ROM MONITOR JUNE 78
SINGLE STEP- WAVEFORMS



New Bear Computing Store
© 1979 ISS. I



FIG. 4

MONITOR PIN OUTS

EDGE CONNECTOR	F	FUNCTIONS	INPUT/ OUTPUT
1	Power	0V	
2	Power	0V	
3	Power	5V	
4	R/W		Input
5	256 SEL	Selects Mikbug Ram on CPU Board	Output
6	VMA		Input
7 Slot			
8	RMD	Disables Data Bus Drivers	Input
9 Sp			
10	Ø2		Input
11	IRQ	From PIA & ACIA	Output
12	NMI	Used in single step function	Output
13	ESS	Enables Single Step function	Output
14	PA7	MIKBUG input port	Input
15	CB	MIKBUG control output	Output
16	PBO	MIKBUG output port	Output
17 Sp			
18 Sp			
19 Sp			
20	DCD	ACIA	
21/36	A0-A15	Address Bus	Inputs
37	0V	Eth. for V24 Driver	
38/45	D7-D0	Data Bus	Input/ output

ROM MONITOR PIN OUTS

EDGE CONNECTOR		Functions	
46	CA1		
47	CA2		
48	PA0		
49	PA1		
50	PA2		
51	PA3		
52	PA4	User	
53	PA5	PIA	
54	PA6		
55	PA7		
56	PB0		
57	PB1		
58	PB2		
59	PB3		
60	PB4		
61	PB5		
62	PB6		
63	PB7		
64	CB1		
65	CB2		
66	RTS	ACIA	
67	TXCLK	ACIA	
68	Spare		
69	RXCLK	ACIA	
70	CTS	ACIA	
71	RX DATA	ACIA	
72	TX DATA	ACIA	
73	RESET	RESETS PIA	Input
74	+12v	Power	
75	-12v	Power	
76	+5v	Power	
77-68	0V	Power	

ROM MONITOR 16/6/78

ROM MEMORY MAP

DEC	Switch Control.	Programme Control	DATA	HEX	Quadrant No
		A2 A1 A0	D7 D6 D5 D4 D3 D2 D1 D0		
0	0 0	0 0 0			
1	0 0	0 0 1			
2	0 0	0 1 0			
3	0 0	0 1 1			1
4	0 0	1 0 0			
5	0 0	1 0 1			
6	0 0	1 1 0	1 0 1 0 0 0 0 0	A0	
7	0 0	1 1 1	0 1 0 0 0 0 0 0	4A	
8	0 1	0 0 0	1 1 1 0 0 0 0 0	E0	
9	0 1	0 0 1	0 0 0 0 0 0 0 0	00	
10	0 1	0 1 0	1 1 1 0 0 0 0 1	E1	
11	0 1	0 1 1	1 0 0 0 1 0 1 1	8B	2
12	0 1	1 0 0	1 1 1 0 0 0 0 1	E1	
13	0 1	1 0 1	1 0 1 0 0 1 1 1	A7	
14	0 1	1 1 0	1 1 1 0 0 0 0 0	E0	
15	0 1	1 1 1	1 1 0 1 0 0 0 0	DO	
16	1 0	0 0 0			
17	1 0	0 0 1			
18	1 0	0 1 0			
19	1 0	0 1 1			3
20	1 0	1 0 0			
21	1 0	1 0 1			
22	1 0	1 1 0			
23	1 0	1 1 1			
24	1 1	0 0 0	1 1 1 0 0 0 0 0	E0	
25	1 1	0 0 1	0 0 0 0 0 0 0 0	00	
26	1 1	0 1 0	1 1 1 0 0 0 0 1	E1	
27	1 1	0 1 1	0 0 0 1 0 0 1 1	13	4
28	1 1	1 0 0	1 1 1 0 0 0 0 0	E0	
29	1 1	1 0 1	0 0 0 0 0 1 0 1	05	
30	1 1	1 1 0	1 1 1 0 0 0 0 0	E0	
31	1 1	1 1 1	1 1 0 1 0 0 0 0	DO	
in nos	14 13	12 11 10	9 7 6 5 4 3 2 1		
82S123					

FIGURE 7

ROM MONITOR 77-68

PARTS LIST

X1	74LS08
X2, 5, 13, 14	DM81LS97
X3	6830L7/SWATBUG
X4, 8	6820
X7	TMS 2708
X15	6850
X6, 9	74LS04
X10, 12	74LS10
X11, 26	74LS127
X16, 22, 23, 27	74LS00
X25, 21, 17	74LS02
X28	74LS163
X30, 29	74LS74
X24	MC14536
X18	MC1488
X19	MC1489
X20	82S123 (ready programmed)
C1, 4	Electrolytic
C2, 3, 5, 6, 7, 8, 9, 11, 13, 14, 15, 16	Ceramic Decoupling
C12	Polystyrene

COMPONENTS LIST

Capacitors

C1, 4	100uf 10v ±18.5mm)
C2, 3 5/11, 13/16	0.1uf Disc Ceramic
C12	470pf

Resistors

R1, 2, 12, 13, 9, 10, 11	10k $\frac{1}{4}$ watt (10mm)
R3/8, 15, 19	2.7 $\frac{1}{4}$ watt
R14	100 $\frac{1}{4}$ watt
R16, 17, 18	10k preset cermet (0.75")

D1	Zener 5.6v
TR1	BC107
TR2	BCY71

Switches

X31	8 SPST Dual in line type
1 off	Single pole toggle switch
1 off	Double pole through toggle switch

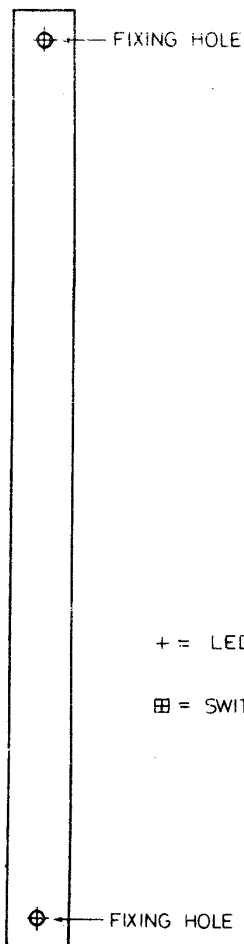
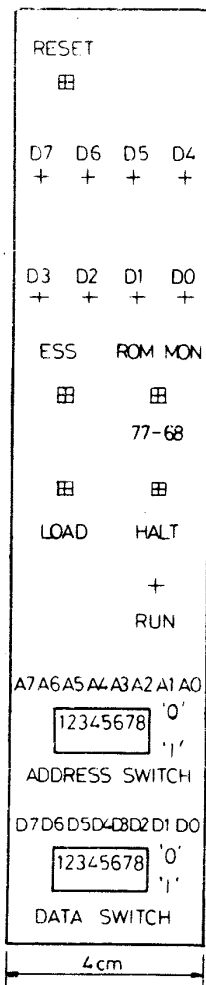
Sockets

2 x 40 pin
3 x 24 pin
1 x 16 pin (ROM)

Wire - 8 yards of 28 SWG

8 inches of 8 way ribbon cable

DRAWING No.



+ = LED's

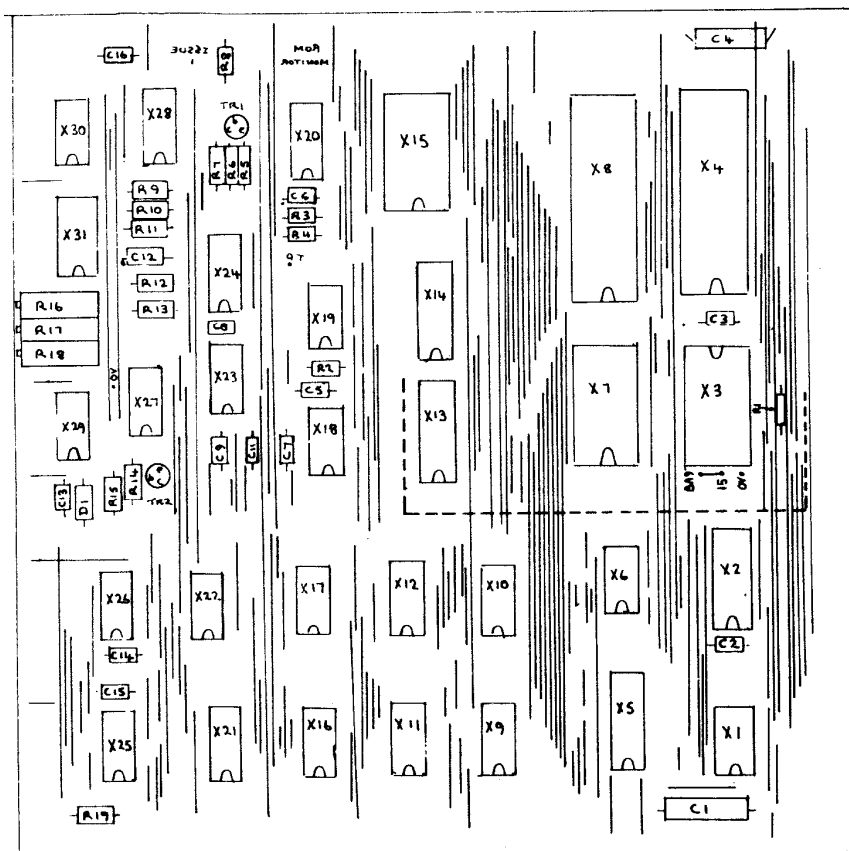
田 = SWITCHES

SCALE 1:1

FIGURE 9

© 1978
NEW BEAR
COMPUTING STORE

DRAWN Ac	CHECKED	APPROVED
DATE 11-8-78	ISSUE	
TITLE		DRAWING No.



EDGE
CONNECTOR

77-68 ROM MON 2
LAYOUT DIAGRAM